MIL-P-60316A(MU)
15 June 1968
SUPERSEDING
MIL-P-60316(MU)
29 January 1965

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M32
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of percussion primer designated as Primer, Percussion, M32.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Method of
MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of
MIL-P-46425	- Primer, Percussion, M35, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

ARMY

8880637 - Primer, Percussion M32 Assembly and Details
9205628 - Box Set-Up, Packing, Ammunition for Primer Percussion M32
9205629 - Box Packing, Ammunition for Primer Percussion M32

PUBLICATIONS

ARMY

ET-8880637 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer.-The primer shall comply with all requirements specified on Drawing (dwg.) 8880637 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder used in loading the primer at the time of loading, shall not exceed 0.5 percent when tested as specified in 4.3.3.

3.4 Sensitivity

3.4.1 Non-functioning.-The primer shall not function under an applied energy value of 8.65 inch ounces minimum (min.) when tested as specified in 4.3.1.1.

3.4.2 Functioning.-The primer shall function under an applied energy value of 82.60 inch ounces maximum (max.) when tested as specified in 4.3.1.2.

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3.5 Proving ground.-The primer shall function on the first attempt, there shall be no audible hangfire and no part shall show evidence of rupture, break or become detached when tested as specified in 4.3.2.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, corrosion and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to act in his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of

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the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Parts from the same lot interfix number from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. Percussion elements from not more than one lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Assembly, prior to assembling closing disc and head (see dwg. 8880637).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Flash hole missing or obstructed	Visual	01001
Major:	AQL 1.0 percent		
101.	Weight of pellet	Balance	01002
102.	Pellet missing	Visual	01003
103.	Percussion element missing	Visual	01004
104.	Percussion element inverted	Visual	01005
105.	Lacquer seal missing or incomplete around percussion element	Visual	01006
106.	Percussion element loose or damaged	Visual-Manual	01007
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	01008

4.2.2.2 Assembly (see dwg. 8880637).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.0 percent		
101.	Firing plug above flush, max.	Gage	02001
102.	Length, max.	Gage	02002
103.	Crimp of housing missing or not 360 degrees	Visual	02003
104.	Closing disc punctured, torn or coating missing	Visual	02004
105.	Seal of closing disc missing or insufficient	Visual	02005
106.	Head assembly loose	Manual	02006
107.	Pitch diameter of thread, min.	Gage	02007
108.	Major diameter of thread, min.	Gage	02008

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Minor:	AQL 1.5 percent		
201.	Marking incorrect or incomplete	Visual	02009
202.	Evidence of poor workmanship (see 3.6)	Visual	02010
203.	Pitch diameter of thread, max.	Gage	02011
204.	Major diameter of thread, max.	Gage	02012

4.2.2.3 Set-up box, prior to sealing (see dwg. 9205628).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Primer missing	Visual	03001
202.	Primer inverted	Visual	03002
203.	Any packing material missing .	Visual	03003

4.2.2.4 Sealed set-up box (see dwg. 9205628).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Sealing strip torn, incomplete, badly wrinkled or otherwise fails to seal box	Visual	04001
202.	Set-up box torn, cut or punctured	Visual	04002
203.	marking missing, misleading or unidentifiable	Visual	04003
204.	Contents loose	Manual	04004

4.2.2.5 Sealed barrier bag (see dwg. 9205628) (as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.4 percent		
101.	Bag, punctured, torn or cut ..	Visual	05001
102.	Seal mislocated or incomplete.	Visual	05002
Minor:	AQL 1.0 percent		
201.	Marking misleading, missing or unidentifiable	Visual	05003
202.	Evidence of poor workmanship (see 3.6)	Visual	05004

4.2.2.6 Packing box (dwg. 9205629).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Protruding nail	Visual	06001
102.	Board broken or loose	Visual-Manual	06002
103.	Hardware strapping missing, broken or loose	Visual-Manual	06003
Minor:	AQL 1.5 percent		
201.	Metallic seal missing, unsealed or improperly positioned	Visual	06004
202.	Hardware or strapping improperly engaged or mislocated	Visual	06005
203.	Marking misleading or unidentifiable	Visual	06006
204.	Contents loose	Manual	06007
205.	Evidence of poor workmanship (see 3.6)	Visual	06008

4.2.3 Testing

4.2.3.1 Moisture content of black powder (see 3.3) - Major defect - Code No. 07001.-A sample of twenty-five pellets of black powder prior to assembly to the primer housing shall be randomly selected from each eight hours production and shall be tested as specified in 4.3.3. Any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, primers loaded with the non-conforming black powder shall be rejected.

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4.2.3.2 Sensitivity

4.2.3.2.1 Non-functioning (see 3.4.1) - Critical defect - Code No. 02013.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified 200 primers shall be randomly selected from each lot for this test. If one or more primers fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.

4.2.3.2.1.1 After three consecutive lots have met the criteria of 4.2.3.2.1, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter K and AQL of 0.10 percent. The test shall be performed as specified in 4.3.1.1.

4.2.3.2.2 Functioning (see 3.4.2) - Major defect - Code No. 02014.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified 315 primers shall be randomly selected from each lot for this test. If 4 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 2 but less than 4 defectives are found a second sample of 315 primers shall be tested. If the combined number of defectives in both the first and second sample is 5 or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2.

4.2.3.2.2.1 After three consecutive lots have met the criteria of 4.2.3.2.2, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter L and AQL of .25 percent. The test shall be performed as specified in 4.3.1.2.

4.2.3.3 Proving ground test (see 3.5) - Major defect - Code No. 07002.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 125 primers shall be selected from each lot for this test. If 2 or more primers fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

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4.2.3.3.1 After three consecutive lots have met the criteria of 4.2.3.3, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter H and AQL of .25 percent. The test shall be performed as specified in 4.3.2.

4.2.3.4 Check test for deterioration of Primer, Percussion (M35).-If the total elapsed time between original acceptance of the M35 primer lot and the assembly of that lot into the M32 primer exceeds two years, or if the M35 primers have been subjected to adverse conditions for any period of time since previous test, the M35 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46425 immediately before the M35 primer lot is assembled into the M32 primers.

4.2.3.5 Heat seal test of barrier bag (see dwg. 9205628) - Major defect - Code No. 05005.-Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-8880037 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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4.3 Test methods and procedures

4.3.1 Sensitivity

4.3.1.1 Non-functioning.-Prior to performing this test the closing disc and black powder pellet shall be removed from the primer. The primer shall then be assembled to the applicable test fixture specified in 4.2.4. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A weight of 2.95 pound minimum shall be dropped on the firing pin from a height of 3/16 inch, minimum, to function the primer. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which functions shall be classed defective. Primers used in this test shall not be returned to the lot.

4.3.1.2 Functioning.-Prior to performing this test the closing disc and black powder pellet shall be removed from the primer. The primer shall then be assembled to the applicable test fixture specified in 4.2.4. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug. A weight of 2.95 pound, maximum, shall be dropped on the firing pin to function the primer from a height of 1 3/4 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which fails to function shall be classed defective.

4.3.2 Proving ground.-This test shall be performed at a Government proving ground. The sample primers shall be assembled with ignition cartridges and dummy fuzes to projectiles for which the primer was designed and which are inert loaded to a total individual weight of approximately 2.90 pounds. One half of the projectiles shall be drop fired and the remaining half shall be trigger fired in mortars for which they are standard. Mortar shall be elevated to an angle of 45 degrees. In lieu of inert loaded projectiles, proof projectiles of acceptable types may be used. Observation shall be made for compliance with the applicable requirements.

4.3.3 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Levels B and C.-The primers shall be preserved and packaged in accordance with dwg. 9205628.

5.2 Packing

5.2.1 Levels B and C.-The containers shall be packed in accordance with dwg. 9205629.

5.3 Marking.-Marking shall be in accordance with dwgs. 9205628 and 9205629.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the LL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Check test for possible deterioration of M32 primers.-The following tests are designed to prevent the loading of major ammunition items with primers which have deteriorated. The tests shall be performed on each lot of primers which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primer is assembled does not require that the check test be performed, the contracting officer shall insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.4.1 Sensitivity

6.4.1.1 Non-functioning.-A sample shall be selected in accordance with Standard MIL-STD-105, Code letter K and an AQL of 0.10 percent. The test shall be performed as specified in 4.3.1.1, using test equipment specified in 4.2.4. Primers used in this test shall not be returned to the lot. Rejection shall be in accordance with Standard MIL-STD-105.

6.4.1.2 Functioning.-A sample shall be selected in accordance with Standard MIL-STD-105, Code letter L and an AQL of 0.50 percent. The test shall be performed as specified in 4.3.1.2, using test equipment specified in 4.2.4, Primers used in this test shall not be returned to the lot. Rejection shall be in accordance with MIL-STD-105.

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6.4.2 Moisture content of black powder.-A sample of 25 primers shall be selected from each lot for this test. The black powder pellet shall be removed from the primer by any suitable means. Since black powder is hygroscopic, it is advisable to place the pellets in a dry, air-tight container immediately upon removal from the primer and to run the determination as soon thereafter as possible. The moisture content of the black powder shall be determined as specified in Specification MIL-P-223. If the moisture content is in excess of 0.7 percent the lot of primers shall be rejected.

6.5 Level B packaging and packing.-In the event the expected storage time of the primers is in excess of 180 days, the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring agency.

6.6 Submission of test data.-In addition to the normal distribution of records, when the primer is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: SMUPA-ND

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-1063

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